

FAREWELL **MiG-21**

IAF's FLYING COFFIN RETIRES



FIRST FLIGHT:
1963

KARGIL WAR
HERO

RETIREMENT:
2025

DEFENCE MATRIX



Cover

MiG-21: End of an Era, Su-57: India's Stealth Gamble, DRDO: Powering Atmanirbharta, Agni-P: Rail-Launched Deterrence, 40N6: Extending Air Defence, AMCA: India's 5th-Gen Future

Foreword

India's defence story is one of courage, innovation, and ambition. From the MiG-21 that soared into history to the cutting-edge AMCA stealth fighter in the making, India has continually pushed the boundaries of aerospace and military technology.

Defence Matrix brings you into the heart of this journey. Inside these pages, you'll find detailed insights into aircraft, missiles, air defence systems, and the genius of India's scientists and engineers. Each story is designed to be easy to read, yet packed with technical depth and strategic perspective.

Discover how the Agni-Prime missile enhances India's second-strike capability. Learn why the S-400 air defence system, with 40N6 missiles, strengthens our borders. Explore the vision behind DRDO's cutting-edge research and the collaborative efforts driving the AMCA program.

This magazine isn't just about machines and technology—it's about India's rise as a self-reliant, modern, and strategically prepared nation. It's about vision, innovation, and the relentless pursuit of excellence.

Dive in, explore, and witness the power, precision, and potential of India's defence capabilities.

Editor's Note

Dear Readers,

When we began shaping *Defence Matrix*, one thought guided us: *India deserves a magazine that speaks about its defence and strategy in a clear, engaging, and accessible way*. Too often, defence discussions are either overly technical or lost in noise. Our goal is to bridge that gap — bringing you credible facts, sharp analysis, and stories that matter.

This first issue is just the beginning. From fighter jets to submarines, from Kargil to the Indo-Pacific, every edition will take you deeper into the world of defence and geopolitics. And most importantly, this is **your magazine**. We want your feedback, your questions, and your curiosity to shape the path ahead.

Together, let's decode, analyse, and understand the forces that protect our nation and shape our future.

Warm regards,

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1. The MiG-21: From Supersonic Marvel to Flying Coffin

“MiG-21 was not just a fighter jet. It was a flying university — shaping generations of IAF pilots, teaching them speed, courage, and survival.”



Few aircraft in aviation history have had a journey as dramatic as the **MiG-21** in India. Born in the Soviet Union in the late 1950s, the delta-winged jet became India's first supersonic fighter and soon evolved into a symbol of national pride. For over six decades, it shaped the Indian Air Force (IAF) — as a war machine, as a training platform, and as a controversial companion.

The Glory Days

When India inducted the MiG-21 in 1963, it revolutionised the IAF. Capable of crossing Mach 2, it gave India a technological edge over Pakistan. The jet played key roles in the **1965 and 1971 wars**, where MiGs defended Indian skies and even turned the tide in several dogfights. In the **1999 Kargil conflict**, upgraded MiG-21 Bisons carried out daring ground-attack missions in high altitude terrain.

The Dark Side

But glory came at a heavy cost. Between 1971 and 2012, more than **480 MiG-21 crashes** were recorded. The aircraft

MiG-21 in Numbers

- **Induction in IAF:** 1963
- **Variants operated by India:** MiG-21FL, M, MF, Bis, Bison
- **Maximum Speed:** Mach 2.05 (~2,175 km/h)
- **Combat Radius:** ~370 km
- **Total built worldwide:** 11,496
- **Total inducted by India:** ~870
- **Crashes in India:** Over 480 (1971–2012)

earned the grim nickname “**flying coffin**”, with many young pilots losing their lives. The reasons were complex: aging design, difficulties sourcing quality spares after the Soviet collapse, and a training system that pushed cadets straight from basic trainers into supersonic MiGs.

Loved & Loathed

Despite its flaws, many veterans defended the MiG-21. They argue that the aircraft delivered far more than its design intended, and blame systemic issues for its poor safety record. For pilots who survived its quirks, the MiG wasn’t just a fighter — it was a teacher.

After the MiG-21: The Future of Indian Air Power

The retirement of the MiG-21 on **26 September 2025** marks not just the end of an era, but also the beginning of new challenges for the IAF. With a sanctioned strength of 42 fighter squadrons but only about 30 operational, the gap is significant.

Quick Takeaways

- **MiG-21** → Iconic but outdated; trained generations of pilots; retired after 62 years.
- **Tejas** → India’s pride; indigenous, cost-effective, still maturing but set to be backbone.
- **Rafale** → Cutting-edge technology; limited numbers but game-changer in deterrence.

What Fills the Void?

- **Tejas Mk1 & Mk2:** Indigenous fighters are expected to form the backbone of the IAF in the coming decades. The government has already cleared procurement of nearly **100 additional Tejas jets**.
- **Rafales & Su-30s:** Together, these two jets currently carry India’s frontline responsibilities — with Rafales adding cutting-edge capability and Su-30MKIs providing versatility.
- **Future Projects:** The upcoming **AMCA (Advanced Medium Combat Aircraft)** will be India’s leap into stealth, while global collaborations may bring in next-generation technologies.

Lessons from the MiG Era

1. **Self-Reliance is Key:** Dependence on foreign spares proved costly in the MiG-21 years.
2. **Safety in Training:** The shift to better trainer aircraft like the Hawk and HTT-40 ensures safer pilot grooming.
3. **Balanced Fleet:** India must ensure numbers, technology, and readiness evolves together — not one at the cost of another.

A Symbolic Moment

Defence Minister Rajnath Singh compared the MiG-21's farewell to ancient Indian rituals where tools were honoured before being set aside. It is a reminder that every machine has a life cycle — but the values of courage, innovation, and adaptability must endure.

Comparison: MiG-21 vs. Tejas vs Rafale

Feature	MiG-21 Bison	HAL Tejas Mk1	Dassault Rafale
Induction (IAF)	1963 (upgraded till 2000s)	2016	2020
Role	Interceptor / Multirole (upgraded)	Light Multirole Fighter	Medium Multirole Fighter
Origin	Soviet Union (built under licence in India)	Indigenous (India)	France
Max Speed	Mach 2.05 (~2,175 km/h)	Mach 1.6 (~2,205 km/h)	Mach 1.8 (~2,222 km/h)
Combat Radius	~370 km	~500 km	~1,850 km
Service Ceiling	58,000 ft	52,000 ft	50,000 ft
Avionics	Upgraded radar & limited BVR	Modern glass cockpit, AESA radar	Advanced AESA radar, SPECTRA EW suite,

Feature	MiG-21 Bison	HAL Tejas Mk1	Dassault Rafale
	(Beyond Visual Range)	(Mk1A onwards)	sensor fusion
Weapons	R-73, R-77 (BVR), unguided bombs	BVR missiles, precision bombs, laser-guided weapons	Meteor BVR, SCALP cruise missile, Hammer bombs, advanced A2A & A2G arsenal

MiG-21 Timeline in India

- **1963:** First MiG-21 inducted into the IAF.
- **1965 War:** MiGs see limited action, establish air defence role.
- **1971 War:** MiG-21s dominate the skies, credited with decisive victories.
- **1980s:** Licence production by HAL peaks, fleet strength at maximum.
- **1999 (Kargil War):** Upgraded MiG-21 Bisons fly precision strikes.
- **2019 (Balakot Crisis):** MiG-21 Bison flown by Wing Commander Abhinandan Varthaman downs a Pakistani F-16.

Stealth Features	None	Some radar cross-section reduction	Partial stealth (reduced RCS)
Numbers in IAF	~870 inducted (phased out 2025)	~40 in service (123+ on order)	36 in service
Notable Combat	1971 war, Kargil, Balakot (2019 dogfight)	Border patrol, frontline deployments	Ladakh standoff (2020), frontline deterrence

Famous MiG-21 Moments

- ✈ **1971 War Heroics** – MiG-21s outclass Pakistan's Sabres & Mirages.
- ✈ **Kargil Ops** – Proved versatile in tough mountain terrain.
- ✈ **Abhinandan's Dogfight (2019)** – A MiG-21 shot down a PAF F-16 in one of the rarest modern aerial encounters.
- ✈ **Flying Coffin Tag** – A painful legacy due to multiple crashes and pilot losses.

2. Su-57 — India's Stealth Leap or Overhyped Interim Fix?

As India continues to modernize its air force, reports suggest the Indian Air Force (IAF) is once again considering acquiring Russia's **Su-57** fighter jet. The proposition isn't just about buying new machines — it's about strategic posture, stealth, strike range, and what role this jet would play in filling gaps until India's indigenous advanced fighters mature.



What is the Su-57?

- The Su-57, Russia's fifth-generation multirole stealth fighter, is intended to combine low observe ability, advanced sensors, super manoeuvrability, and internal weapons carriage.
 - Key features claimed include: AESA radar, advanced electronic warfare (EW) capabilities, ability to carry long-range air-to-air missiles (like the R-37M), internal weapon bays, and enhanced payload capacity when compared to older jets.
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India's Interest: Why Now?

- India is reportedly evaluating acquisition of **two squadrons** of Su-57s, with potential for additional units to be manufactured domestically by HAL in Nashik.
- The interest is driven by shortfalls: current squadron strength has been lagging required numbers; on-going border tensions require long-reach strike capabilities. Su-57's ability for deep strike, SEAD (Suppression of Enemy Air Defences), and BVR (Beyond Visual Range) engagements make it attractive.

Technical Comparison: Su-57 vs. Su-35

Parameter	Su-57	Su-35
Generation & Design Philosophy	5th gen stealth + multirole; emphasis on internal weapon bays and sensors.	4.5-gen; excellent manoeuvrability, proven performance; carries weapons externally
Stealth / Radar Cross Section (RCS)	Lower than Su-35; estimated RCS of 0.1-0.5 m ² . That's a big drop vs typical 4-gen fighter, though far behind American top stealth jets.	Minimal stealth design; large external stores further increase detection risk.
Super cruise & Speed	Claims of super cruise are disputed; its engines (current variants) may not sustain supersonic flight without afterburners over long durations.	High top speed (above Mach 2), excellent thrust; but no stealth in performance.
Combat Radius / Range	Shorter than Su-35 in some operational modes due to stealth and internal weapon/aerodynamic constraints.	Larger combat radius in non-stealth, external load configurations.
Payload Capacity	Internally limited; external stores increase detection. But the internal bay allows stealth missions with missiles/weapons.	Larger external payload possible; more flexibility in heavy ordnance, but at cost of stealth.

Pros and Challenges for India

Pros

1. **Stealth & Sensor Fusion:** Even limited stealth gives India an edge in contested airspace. Paired with modern sensors, the platform improves survivability.
2. **Deep Strike Capability:** With long-range missiles (e.g. R-37M, possible hypersonic weapons), Su-57 could strike deep into adversary territory while engaging or evading air defences.
3. **Interim Fill for AMCA:** India's indigenous fighter, AMCA, is still years away. Su-57 could bridge the gap.
4. **Production & Transfer Opportunities:** If HAL can be involved via licensing or local assembly, not only does India get jets, it builds industrial capability.

Challenges

1. **Stealth Limitations:** Su-57's stealth is often described as "semi-stealth". Some external payloads or pods will compromise its low observability. For true stealth operations, detectability against advanced radars could be higher than ideal.
2. **Engine & Reliability Risks:** Russian engines historically have had issues. Maintenance spares, and performance in varied Indian environmental conditions (heat, humidity, high altitude) need careful evaluation.
3. **Cost vs Benefit:** Importing or even partially local-assembling high-technology aircraft is expensive. Costs of infrastructure, training, maintenance are high.
4. **Doctrinal Fit:** Is it better used like the Su-30MKI (heavy payload + external stores) or as stealthy penetrator? The chosen role affects how useful it will be and how to invest in it.

In Indian Context: What Role Could It Play?

- **SEAD & Deep Strike Missions:** Particularly along northern and northeaster borders, where enemy air defences are robust, a stealth-capable aircraft helps neutralize radar/missile threats.
- **Countering Chinese Stealth Jets:** China's J-20 force is growing; getting Su-57 will give India a more symmetric tool in stealth or stealth-aware warfare.

- **Supporting Regional Power Projection:** For deterrence and regional influence, long-range strike from Su-57s will send strong signals.
 - **Bridging Capabilities Until AMCA:** The AMCA is projected to have stealth, advanced sensors, local manufacture; but until that is ready and reliable, Su-57 could keep India's air force modern and relevant.
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Conclusion: Stealth or Strike? Or Both

India's interest in Su-57 represents a careful balancing act. On one hand, it needs stealth, advanced sensors, internal weapon bays — qualities that define modern air superiority. On the other, practical needs (payload, range, cost, readiness) often push aircraft toward more versatile, but less stealthy configurations.

If India acquires Su-57s, the ideal path might be:

- Use them **predominantly in stealth mode** with internal weapons for high-risk missions;
- Leverage external payloads only when stealth is less crucial;
- Invest heavily in sensors, pilot training, maintenance infrastructure;
- And integrate the Su-57 fleet closely with other platforms (Su-30MKI, Tejas, Rifles) so that the strengths of each cover the weaknesses of others.

In short: the Su-57 could be a game changer *if* India treats it not just as another fighter, but as a stealthy long-strike platform that elevates the entire force structure, rather than merely “more aircraft”.

Jet Specs at a Glance

Fighter Jet	Origin	Generation	Max Speed	Combat Radius	Payload	Stealth / RCS	Key Role
Su-57 (E)	Russia	5th Gen (Semi-Stealth)	Mach 2.0	~1,500 km	~10,000 kg (internal + external)	Reduced RCS (0.1–0.5 m² est.)	Stealth strike, SEAD, BVR combat
Su-30MKI	India/Russia	4.5 Gen	Mach 2.0	~1,500–1,800 km	~8,000 kg	Large RCS (not stealth)	Heavy payload, multi-role, long-range
Tejas Mk-1A	India	4+ Gen	Mach 1.6	~500 km	~3,500–4,000 kg	Medium RCS	Light fighter, short-range strike, air defence
Rafale (F3R)	France	4.5 Gen	Mach 1.8	~1,000–1,100 km	~9,500 kg	Moderate RCS (stealth coatings)	Precision strike, nuclear delivery, multi-role
J-20 (Mighty Dragon)	China	5th Gen Stealth	Mach 2.0	~1,100–1,200 km	~11,000 kg	Low RCS (est. 0.05–0.1 m²)	Stealth penetration, BVR air dominance

3. Forging India's Defence Future

The Defence Research and Development Organisation (DRDO) have undergone a remarkable transformation, evolving from a struggling research entity to a cornerstone of India's defence infrastructure. Overcoming persistent denial regimes, international sanctions, and strict technology controls, DRDO's progress has been propelled by the nation's resolve and the dedication of its scientists and industry partners.



A Legacy of Self-Reliance

India's journey towards defence self-reliance has been marked by significant milestones. The successful development and deployment of indigenous systems like the Agni and Prithvi missile series, the Light Combat Aircraft Tejas, the Akash air defence system, and the aircraft carrier INS Vikrant have showcased DRDO's capabilities. These achievements have not only enhanced India's defence capabilities but also bolstered national pride and confidence.

Strategic Collaborations and Technology Transfers

DRDO has actively engaged in collaborations with various industry partners to scale up and commercialize advanced materials for both strategic and industrial

use. The successful transfer of over 2,000 licensing agreements and more than 200 production licenses in 2024 underscores DRDO's commitment to building a robust indigenous defence industrial base.

Embracing Innovation and Future Technologies

Under the leadership of DRDO Chairman Dr. Samir V. Kamat, the organization is focusing on developing future-ready defence technologies. Initiatives like Project Kusha aim to bridge gaps in air defence capabilities, while advancements in hypersonic, directed energy and electronic warfare are positioning India at the forefront of modern warfare technologies.

Educational Initiatives and Capacity Building

Recognizing the importance of skilled professionals in the defence sector, DRDO has partnered with institutions like Symbiosis Skills and Professional University in Pune to launch a Bachelor of Technology program in Defence Technology. This initiative aims to address the increasing demand for skilled professionals and foster innovation in defence technologies. [The Times of India](#)

Conclusion

DRDO's journey from a nascent research organization to a global leader in defence technology is a testament to India's commitment to self-reliance and innovation. As the organization continues to evolve, its focus on strategic collaborations, technological advancements, and capacity building will play a pivotal role in shaping India's defence future.

4. Agni-Prime – India's Strategic Leap in Mobile Nuclear Deterrence

On September 25, 2025, India achieved a significant milestone in its strategic defence capabilities by successfully test-firing the Agni-Prime missile from a rail-based mobile launcher. This development marks India as one of the select nations possessing the capability to launch long-range ballistic missiles from rail platforms, joining the ranks of the United States, Russia, China, and possibly North Korea.



Agni-Prime: A Technological Overview

The Agni-Prime (Agni-P) is an intermediate-range ballistic missile (IRBM) with a range of up to 2,000 kilometres, designed to deliver a 1.5-ton payload. It is a two-stage, solid-propellant missile utilizing advanced composite materials, which reduce its weight and enhance its mobility. The missile features a cold-launch system, allowing it to be fired from a sealed canister, which is crucial for rapid deployment and protection against environmental factors.

Equipped with a dual redundant navigation and guidance system, the Agni-Prime ensures high accuracy. It employs an advanced ring laser gyro-based inertial navigation system (INS) and a micro inertial navigation system (MINS), augmented by satellite navigation systems like GPS and India's own NavIC. The

missile can be fitted with a Manoeuvrable Re-entry Vehicle (MaRV), enhancing its capability to penetrate enemy missile defence systems

Strategic Significance of Rail-Based Launch Capability

The integration of the Agni-Prime missile with a rail-based mobile launcher introduces several strategic advantages:

- **Enhanced Mobility:** Rail networks, spanning approximately 70,000 kilometres across India, provide extensive coverage, enabling the missile to be deployed rapidly across the country without the need for extensive infrastructure modifications.
 - **Reduced Vulnerability:** Launching from a moving platform makes it challenging for adversaries to predict and target missile locations, thereby enhancing the survivability of India's nuclear arsenal.
 - **Quick Reaction Capability:** The rail-based system allows for swift deployment and launch, crucial for a credible second-strike capability under India's "No First Use" nuclear doctrine.
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Comparison with Agni-5 and Strategic Positioning

While the Agni-Prime offers enhanced mobility and a shorter range suitable for regional deterrence, the Agni-5 is an intercontinental ballistic missile (ICBM) with a range exceeding 5,000 kilometres, designed for global strategic deterrence. The Agni-Prime complements the Agni-5 by providing a flexible and mobile option for regional defence scenarios, ensuring a comprehensive and credible deterrence posture

Conclusion

The successful rail-based launch of the Agni-Prime missile signifies a leap forward in India's strategic defence capabilities. By combining advanced missile technology with mobile launch platforms, India enhances its deterrence posture, ensuring a credible and survivable second-strike capability. This development underscores India's commitment to strengthening its defence infrastructure and maintaining peace and stability in the region.

5. India Eyes Additional 40N6 Missiles from Russia to Bolster Air Defence Arsenal

In a strategic move to enhance its air defence capabilities, India is reportedly planning to procure additional units of the advanced 40N6 ultra-long-range surface-to-air missile (SAM) from Russia. This acquisition aims to augment the Indian Air Force's (IAF) S-400 Triumph systems, providing extended coverage against a spectrum of aerial threats amid evolving regional security dynamics.

40N6 Missile: Technical Overview

The 40N6 missile is a pivotal component of the S-400 Triumph air defence system, renowned for its impressive engagement range of up to 400 kilometres. Designed to intercept a wide array of targets—including fighter aircraft, unmanned aerial vehicles (UAVs), cruise missiles, and even ballistic and hypersonic threats—it employs active radar homing for precision guidance. Its capability to operate over the horizon, leveraging data from ground-based radars or airborne assets, allows for deep-strike interdiction far beyond traditional SAM envelopes.



The missile's two-stage propulsion system enables it to reach altitudes exceeding 30 kilometres, enhancing its effectiveness against high-altitude

targets. Its high velocity and manoeuvrability make it a formidable asset in modern air defence operations.

Strategic Significance for India

India's interest in acquiring additional 40N6 missiles underscores its commitment to strengthening its air defence infrastructure. The planned procurement is set to enhance the operational reach of the S-400 systems, particularly along India's borders with Pakistan and China, where air superiority remains a critical factor. This move aligns with India's broader defence strategy to bolster its deterrence capabilities and maintain a technological edge in the region.

Operational Performance: A Case Study

The real-world efficacy of the 40N6 missile was demonstrated during a military engagement between India and Pakistan in May 2025, referred to as "Operation Sindoor." During this operation, the missile was employed to target a Pakistan Air Force (PAF) Intelligence, Surveillance, and Reconnaissance (ISR) aircraft at an unprecedented distance of 314 kilometers. This engagement marked a significant milestone in demonstrating the missile's long-range interception capabilities.

The successful interception disrupted Pakistani aerial operations, contributing to India's tactical advantage during the conflict. This incident highlighted the strategic value of the 40N6 missile in modern warfare scenarios.

Conclusion

India's pursuit of additional 40N6 missiles from Russia represents a strategic enhancement of its air defence capabilities. By augmenting the S-400 Triumph systems with these advanced missiles, India aims to fortify its defence posture and ensure robust deterrence against a spectrum of aerial threats. This development underscores the evolving dynamics of India's defence strategy in response to regional security challenges.

6. India's AMCA Fighter Jet Programme – A Strategic Leap in Indigenous Defence

India's ambition to develop a fifth-generation stealth fighter jet, the Advanced Medium Combat Aircraft (AMCA), has gained significant momentum with recent strategic partnerships. Larsen & Toubro (L&T) and Bharat Electronics Limited (BEL) have joined forces to bolster the AMCA programme, aiming to deliver a multi-role, stealth-capable aircraft for both the Indian Air Force (IAF) and Navy.



Strategic Partnerships: L&T and BEL

The collaboration between L&T and BEL combines L&T's expertise in strategic defence and aerospace platforms with BEL's strengths in defence electronics and systems. This partnership builds upon their previous involvement in the indigenous Light Combat Aircraft (LCA) programme, where they supplied major aero structure modules and developed mission-critical avionics and electronic systems. The consortium plans to respond to the Expression of Interest (EoI) issued by the Government of India's Aeronautical Development Agency in the coming weeks.

AMCA: Design and Specifications

The AMCA is envisioned as a single-seat, twin-engine, all-weather multirole stealth combat aircraft. Key design features include:

- **Stealth Design:** Incorporation of stealth technologies to reduce radar cross-section.
- **Internal Weapons Bay:** Capacity of 1,500 kg to house precision-guided munitions.
- **External Payload Capacity:** 5,500 kg for additional weaponry.
- **Fuel Capacity:** 6,500 kg to support extended range operations.
- **Maximum Take-off Weight:** Approximately 25 tonnes.
- **Operational Altitude:** Up to 55,000 feet.

The prototype rollout is targeted between 2028 and 2029, with series production expected by 2032–33 and induction into the IAF scheduled for 2034.

Collaborative Efforts: Bharat Forge, BEML, and Data Patterns

In parallel, Bharat Forge Limited (BFL), Bharat Earth Movers Limited (BEML), and Data Patterns have joined hands to bid for the AMCA programme. This consortium aims to contribute to the indigenous manufacturing efforts, particularly in areas such as:

- **Airframe Structures:** Design and fabrication of lightweight and durable structures.
- **Avionics Systems:** Development of advanced avionics for enhanced situational awareness.
- **Mission Systems Integration:** Ensuring seamless integration of various systems for optimal performance.

This collaboration underscores India's commitment to self-reliance in defence manufacturing and the development of indigenous technologies.

Comparison: AMCA vs. Tejas Mk-2

Feature	AMCA	Tejas Mk-2
Role	Stealth Multirole Fighter	Light Multirole Fighter
Generation	5th Gen	4.5th Gen
Engines	Twin-Engine	Single-Engine
Stealth Features	Advanced Stealth Design	Limited Stealth Features
Payload Capacity	1,500 kg (Internal), 5,500 kg (External)	1,000 kg (Internal), 3,500 kg (External)
Operational Altitude	Up to 55,000 feet	Up to 50,000 feet
Induction Timeline	2034	2026

The AMCA is designed to complement the Tejas Mk-2, providing the IAF with a versatile and potent mix of capabilities.

Conclusion

The strategic partnerships between L&T, BEL, BFL, BEML, and Data Patterns signify a concerted effort towards achieving self-reliance in defence manufacturing. The AMCA programme represents a significant leap in India's capabilities, aiming to deliver a world-class fifth-generation fighter aircraft that meets the evolving needs of the Indian Armed Forces.

Defence Matrix Quiz – Test Your Knowledge

1. The MiG-21 was inducted into the Indian Air Force in which year?

- A) 1959
- B) 1965
- C) 1971
- D) 1980

2. Which stealth fighter is India planning to acquire from Russia?

- A) Su-30MKI
- B) Su-35
- C) Su-57
- D) J-20

3. DRDO's Project Kusha focuses on which area of technology?

- A) Light combat aircraft
- B) Air defence missile systems
- C) Airborne hypersonic and next-gen weapons
- D) Naval destroyers

4. What is the maximum range of the Agni-Prime missile?

- A) 500 km
- B) 1,000 km
- C) 1,500 km
- D) 2,000 km

5. The 40N6 missile is part of which Indian air defence system?

- A) Akash
- B) Barak-8
- C) S-400 Triumph
- D) Nag Missile

6. AMCA stands for:

- A) Advanced Medium Combat Aircraft
- B) Armed Multirole Combat Aircraft
- C) Advanced Missile Carrier Aircraft
- D) Airborne Multirole Combat Aviation

7. Which consortium recently partnered for India's AMCA program?

- A) L&T + BEL
- B) Bharat Forge + BEML + Data Patterns
- C) Both A and B
- D) HAL + DRDO

8. The MiG-21 was often called:

- A) Flying Fortress
- B) Flying Coffin
- C) Sky King
- D) Iron Eagle

9. Which Indian missile system enhances second-strike capability via rail-based launchers?

- A) Agni-I
- B) Agni-III
- C) Agni-Prime
- D) Prithvi-II

10. Which of the following is NOT a feature of the AMCA fighter jet?

- A) Twin-engine configuration
- B) Advanced stealth design
- C) Operational altitude up to 55,000 feet
- D) Intercontinental ballistic missile payload

Defence Matrix Quiz – Answers & Explanations

1. The MiG-21 was inducted into the Indian Air Force in which year?

Answer: A) 1959

Explanation: The MiG-21 entered IAF service in 1959 and served for over six decades, becoming an iconic fighter despite its high accident rate.

2. Which stealth fighter is India planning to acquire from Russia?

Answer: C) Su-57

Explanation: India is considering acquiring the Russian Su-57 stealth fighter to enhance its fifth-generation fighter capabilities.

3. DRDO's Project Kusha focuses on which area of technology?

Answer: C) Airborne hypersonic and next-gen weapons

Explanation: Project Kusha is a DRDO initiative aimed at developing advanced air defence technologies, including hypersonic and directed-energy systems.

4. What is the maximum range of the Agni-Prime missile?

Answer: D) 2,000 km

Explanation: Agni-Prime is an intermediate-range ballistic missile with a range of approximately 2,000 km, capable of carrying nuclear payloads and designed for mobility.

5. The 40N6 missile is part of which Indian air defence system?

Answer: C) S-400 Triumph

Explanation: The 40N6 is an ultra-long-range surface-to-air missile integrated into India's S-400 air defence system.

6. AMCA stands for:

Answer: A) Advanced Medium Combat Aircraft

Explanation: AMCA is India's indigenous fifth-generation fighter jet program, aiming for stealth, multi-role capability, and advanced avionics.

7. Which consortium recently partnered for India's AMCA program?

Answer: C) Both A and B

Explanation: L&T with BEL and Bharat Forge + BEML + Data Patterns are all key collaborators in the AMCA program, covering airframe, avionics, and system integration.

8. The MiG-21 was often called:

Answer: B) Flying Coffin

Explanation: Due to its high accident rate over decades of service, the MiG-21 earned the nickname "Flying Coffin," despite its combat success.

9. Which Indian missile system enhances second-strike capability via rail-based launchers?

Answer: C) Agni-Prime

Explanation: Agni-Prime can be launched from rail-mobile platforms, improving survivability and second-strike capability under India's "No First Use" nuclear doctrine.

10. Which of the following is NOT a feature of the AMCA fighter jet?

Answer: D) Intercontinental ballistic missile payload

Explanation: AMCA is a fifth-generation stealth fighter with twin engines and advanced avionics, but it does not carry ICBM payloads; it is an air combat aircraft, not a missile system.

TWO WORD

India's journey in defence technology is a story of vision, determination, and resilience. From the legendary MiG-21 that carved its place in history, to the cutting-edge AMCA stealth fighter and rail-launched Agni-Prime missiles, every program reflects the ingenuity, skill, and dedication of our scientists, engineers, and armed forces.

As you turn the last page of this edition, we hope you feel the pulse of India's defence evolution—the pride of a nation that continues to rise in strategic strength, technological prowess, and self-reliance. Each aircraft, missile, and system is not just hardware; it is a symbol of India's commitment to peace through preparedness, and of the brave men and women who safeguard our skies, seas, and borders.

But the journey doesn't stop here. In the next edition of *Defence Matrix*, we will explore:

- India's naval modernization, including the latest submarine and aircraft carrier programs.
- Cutting-edge drone and unmanned systems shaping modern warfare.
- Indigenous missile defence innovations beyond the S-400.
- Special insights into how India is preparing for future aerial combat and next-generation threats.

Stay with us as we continue to chart India's rise as a modern, strategically capable, and technologically advanced nation. The next chapter promises even more stories, innovations, and breakthroughs that every defence enthusiast, student, and professional will want to explore.

India's defence is our pride, and *Defence Matrix* is here to bring you closer to the heart of every achievement.